

CHAPTER 18

PRACTICAL

ELECTRICITY

4.4 Practical electricity

4.4.1 Uses of electricity

- 1 State common uses of electricity, including heating, lighting, battery charging and powering motors and electronic systems
- 2 State the advantages of connecting lamps in parallel in a lighting circuit
- 3 Recall and use the equation
power = current $\times$ voltage
 $P = IV$
- 4 Recall and use the equation
energy = current $\times$ voltage $\times$ time
 $E = IVt$
- 5 Define the kilowatt-hour (kWh) and calculate the cost of using electrical appliances where the energy unit is the kWh

4.4.2 Electrical safety

- 1 State the hazards of:
 - (a) damaged insulation
 - (b) overheating cables
 - (c) damp conditions
 - (d) excess current from overloading of plugs, extension leads, single and multiple sockets when using a mains supply
- 2 Explain the use and operation of trip switches and fuses and choose appropriate fuse ratings and trip switch settings
- 3 Explain what happens when a live wire touches a metal case that is earthed
- 4 Explain why the outer casing of an electrical appliance must be either non-conducting (double-insulated) or earthed
- 5 Know that a mains circuit consists of a live wire (line wire), a neutral wire and an earth wire and explain why a switch must be connected into the live wire for the circuit to be switched off safely
- 6 Explain why fuses and circuit breakers are connected into the live wire

1. O/N/23/11/Q28

Three students make comments about appliances with double insulation.

student P Appliances with double insulation do not need an earth wire.

student Q Double insulation protects the user of the appliance from an electric shock.

student R The circuits of appliances with double insulation do not contain fuses.

Which students are correct?

- A** P, Q and R **B** P and Q only **C** P and R only **D** Q and R only

2. M/J/23/12/Q31

The power consumption of a microwave oven is 1500 W.

The cost of 1.0 kWh of electrical energy is 24 cents.

What is the cost of using the microwave oven for 10 minutes?

- A** 6 cents
B 36 cents
C 60 cents
D 360 cents

3. M/J/23/11/Q31

Three students each suggest a consequence of wiring lamps in parallel rather than in series.

student 1 If one lamp breaks, the other lamps stay lit.

student 2 The lamps can be switched on and off separately.

student 3 The p.d. across each lamp is equal to the p.d. across the supply.

Which students are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

4. O/N/22/11/Q32

Why is a fuse used in an electrical appliance?

- A** to earth the appliance
B to protect the appliance
C to change the power of the appliance
D to change the voltage supplied to the appliance

5. O/N/22/12/Q33

Which device is used to convert electrical energy to kinetic energy?

- A** iron
- B** kettle
- C** lamp
- D** motor

6. M/J/22/12/Q35

When the flash on a camera is used, a charge of 1.5 C flows for 0.0030 s through the flash lamp.

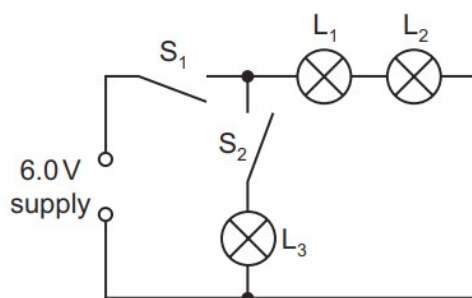
The average voltage across the flash lamp is 3600 V.

What is the electrical energy supplied to the flash lamp and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

7. M/J/22/12/Q36

The diagram shows an electric circuit.

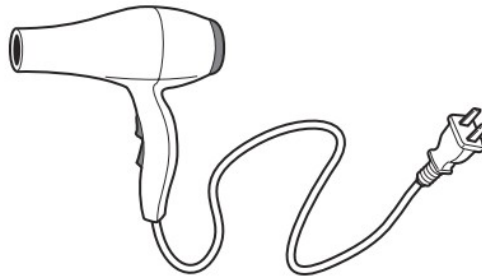


Which lamps light when switch S₁ is closed and S₂ is open?

- A** L₁ only
- B** L₃ only
- C** L₁ and L₂ only
- D** L₁, L₂ and L₃

8. M/J/22/12/Q37

The diagram shows a hairdryer.



Which statement explains why the hairdryer does **not** need an earth wire?

- A It has a live wire.
- B It has a neutral wire.
- C It has double insulation.
- D The cable is insulated.

9. M/J/22/11/Q35

A metal kettle is plugged into a mains socket. The plug contains a suitable fuse.

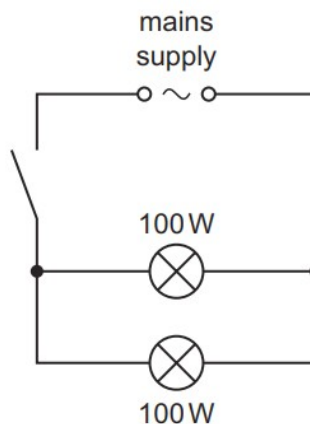
The kettle's cable is damaged and the fuse blows.

Why does the fuse blow?

- A The bare earth wire in the cable touches the bare live wire.
- B The metal casing of the kettle touches the bare earth wire.
- C The bare neutral wire in the cable touches the bare earth wire.
- D The live wire breaks without touching any other wire.

10. O/N/21/11/Q32

Two 100 W lamps are connected in parallel to the mains supply, as shown.



How much electrical energy is supplied by the mains when the switch is closed for 36 minutes?

- A 0.030 kWh
- B 0.060 kWh
- C 0.12 kWh
- D 7.2 kWh

11. O/N/21/11/Q33

The cost of electricity is 9.0 cents/kWh.

It takes 0.060 hours and costs 1.2 cents to boil the water in a kettle.

What is the power rating of the kettle?

- A** 0.45 kW **B** 0.65 kW **C** 2.2 kW **D** 130 kW

12. M/J/21/12/Q35

A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is \$0.24.

How much does it cost to run the lamp for this time?

- A** \$0.36 **B** \$0.84 **C** \$2.52 **D** \$25.20

13. M/J/21/12/Q36

A three-pin mains plug contains a fuse. A device with double insulation has a cable connected to the plug.

Which part of the plug is **not** connected to the cable?

- A** earth pin
B fuse
C live pin
D neutral pin

14. M/J/21/11/Q31

The information on the back of an electric room heater is shown.

rating 220–240 V
~50 Hz
4.2 A

What is a suitable fuse rating for this room heater?

- A** 4.0 A **B** 4.2 A **C** 5.0 A **D** 13.0 A

15. O/N/20/11/Q39

A student calculates the amount of energy used by an electric heater.

What is the equation for calculating the energy E in kWh?

- A** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (second)}$
- B** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (hour)}$
- C** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (second)} \div 1000$
- D** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (hour)} \div 1000$

16. O/N/20/12/Q33

A 240 V mains circuit contains eight 60 W lamps in parallel. At the time when the lamps are switched on, the filaments are cold and the current is four times as large as the final steady current in the circuit.

What is the initial current supplied by the mains?

- A** 0.25 A
- B** 1.0 A
- C** 2.0 A
- D** 8.0 A

17. M/J/20/11/Q40

A kettle is connected to the 240 V mains supply using a plug containing a 13 A fuse.

The kettle contains water.

When it is switched on, the fuse blows. This happens again after a new fuse is fitted.

Someone replaces the fuse with a nail, and the kettle works.

What else might happen as a result of replacing the fuse with a nail?

- A** A very large current overloads the wiring, causing a fire.
- B** The kettle boils the water less quickly.
- C** The kettle uses more energy to boil the water.
- D** The water boils at a higher temperature.

18. O/N/19/11/Q36

A lamp is connected to a mains plug.

Where are the switch and the fuse connected so that the lamp is safe to use?

	switch	fuse
A	live wire	live wire
B	live wire	neutral wire
C	neutral wire	live wire
D	neutral wire	neutral wire

19. O/N/19/12/Q36

A mains electric circuit is fitted with a circuit breaker rather than a fuse.

Where is the circuit breaker connected and what happens when the current is too large?

	A circuit breaker is connected in...	When the current is too large...
A	the live wire.	a thin wire melts and breaks the circuit.
B	the live wire.	an electromagnet opens a switch.
C	the neutral wire.	a thin wire melts and breaks the circuit.
D	the neutral wire.	an electromagnet opens a switch.

20. M/J/19/12/Q35

The cable to an electric cooker contains a live wire, a neutral wire and an earth wire.

When the cooker is working correctly, in which wires are the currents equal?

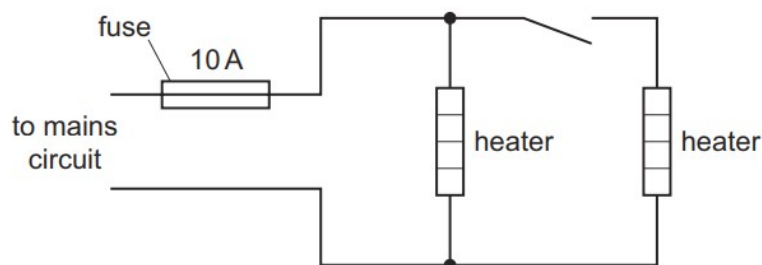
- A** the live, the neutral and the earth
- B** the live and the earth only
- C** the neutral and the earth only
- D** the neutral and the live only

21. M/J/19/12/Q36

The manufacturer of an electrical heater states that only one heater must be connected to the mains wiring through any fuse.

The current in the circuit with only one heater is 9 A. The fuse rating is 10 A.

Despite the warning, a second identical heater is added, as shown. When the switch is closed the fuse blows.



The fuse is replaced with a 20 A fuse.

What happens when the switch is closed?

- A** Each heater only receives half the correct current.
- B** Each heater receives twice the correct current.
- C** The circuit wiring connecting the heaters to the mains may overheat.
- D** The fuse melts when the heaters are switched on.

22. O/N/18/11/Q31

The power of an electrical heater is 3.0 kW.

The heater is switched on for a time of 2 hours and 30 minutes.

The cost of using the heater for this time is 120 c.

What is the cost of 1.0 kWh of electrical energy?

- A** 16 c **B** 17 c **C** 40 c **D** 48 c

23. O/N/18/11/Q32

In a lighting circuit, the switch is placed in the live wire.

Why is this?

- A** A lamp in the circuit can be isolated from the supply.
B The fuse is in the neutral wire.
C The switch does not work in the neutral wire.
D Too much current flows in the earth wire.

24. M/J/18/11/Q31

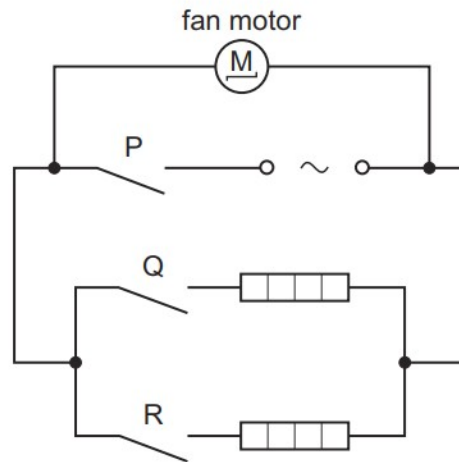
An electrical appliance is plugged into a socket in the wall. The plug contains a fuse.

What is the main purpose of the fuse?

- A** to earth the appliance
B to earth the plug
C to protect the user from electric shock
D to protect the wiring from overheating

25. O/N/17/11/Q32

The diagram shows the circuit for a hair-dryer.



The fan motor has a power rating of 0.10 kW and the heaters each have a rating of 0.40 kW. The cost of electricity is 8.0 cents/kWh.

What is the cost of running the hair-dryer for two hours with switches P and Q closed and switch R open?

- A** 1.6 cents **B** 4.0 cents **C** 6.4 cents **D** 8.0 cents

26. O/N/17/11/Q33

Why are the metal casings of electrical appliances earthed?

- A** to complete the circuit
- B** to ensure the casing is not at a dangerous voltage
- C** to ensure the fuse blows when the current in the appliance is too large
- D** to protect the appliance from overheating

27. O/N/16/12/Q35

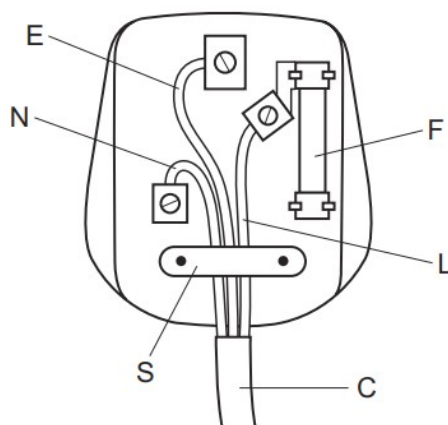
The headlamp circuit in a car includes a 7A fuse.

What is the purpose of the fuse?

- A** It increases the resistance of the circuit.
- B** It makes it easier for the current to flow.
- C** It protects the wiring from overheating.
- D** It reduces the voltage across the lamp.

28. M/J/16/12/Q30

The diagram shows the wiring of a three-pin mains plug. There is an error in the diagram.



What is the error?

- A** The cable cover C is not under the clip S.
- B** The earth wire E is connected to the wrong terminal.
- C** The fuse F is connected to the live wire L.
- D** The live wire L is connected to the wrong end of the fuse F.

29. M/J/16/12/Q31

A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is \$0.24.

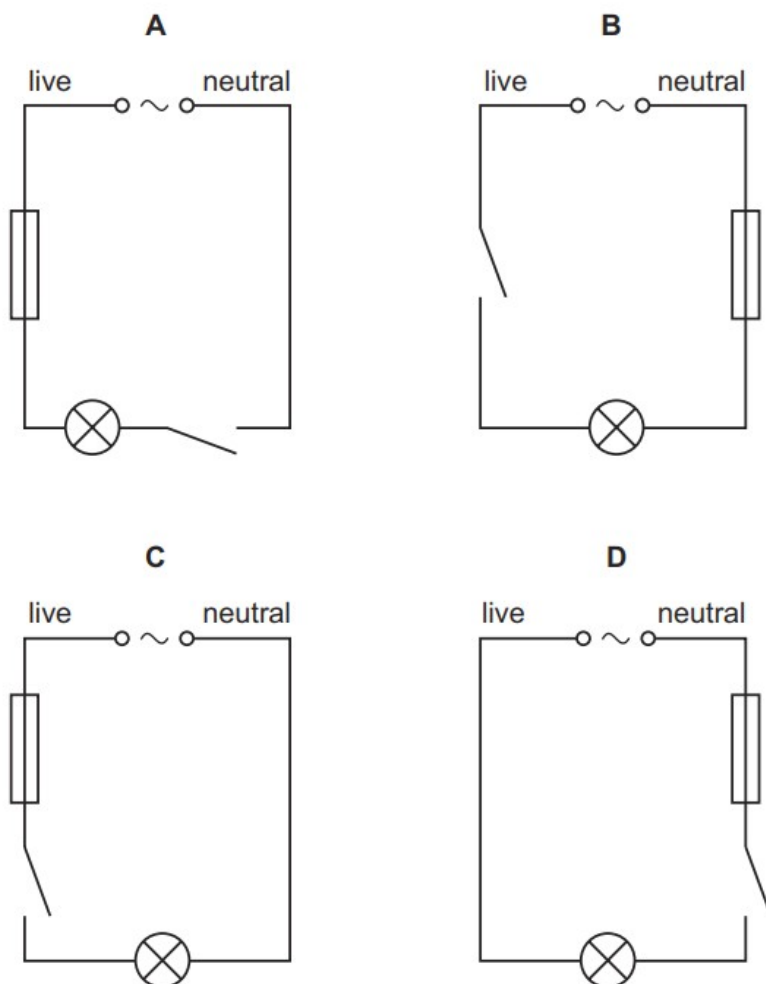
How much does it cost to run the lamp for this time?

- | | | | |
|-----------------|-----------------|-----------------|------------------|
| A \$0.36 | B \$0.84 | C \$2.52 | D \$25.20 |
|-----------------|-----------------|-----------------|------------------|

30. M/J/16/11/Q33

A lamp is connected to the a.c. mains supply in series with a switch and a fuse.

Which circuit shows these components wired correctly?



31. O/N/15/12/Q34

An immersion heater is labelled 24 V, 120 W.

What is the current in the heater when it is connected to a 24 V supply?

- A** 0.20 A **B** 5.0 A **C** 24 A **D** 120 A

32. M/J/15/11/Q31

When the flash on a camera is used, a charge of 1.5 C flows through the flash-tube in 0.0030 s.

The average voltage across the flash-tube is 3600 V.

What is the electrical energy supplied to the flash-tube and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

33. O/N/14/12/Q28

There is a current of 0.25 A in a lamp connected to a 240 V supply.

What is the input power to the lamp?

- A** 15 W **B** 60 W **C** 240 W **D** 960 W

34. O/N/14/12/Q29

Which unit measures the energy input to an electrical appliance?

- A** ampere
B kilowatt-hour
C volt
D watt

35. O/N/14/12/Q30

An electric lamp is marked 0.5 A. It is connected to a socket marked '30 A maximum'.

Which fuse is best to use in the lamp?

- A** 0.5 A **B** 3 A **C** 30 A **D** 40 A

36. M/J/14/12/Q33

An appliance uses a current of 3 A.

Which row is correct for the fuse in this appliance?

	most suitable fuse rating / A	fuse connected in
A	5	earth wire
B	5	live wire
C	13	earth wire
D	13	live wire

37. M/J/14/11/Q32

One kilowatt-hour of energy costs \$0.24.

How much does it cost to run a 2 kW heater for three hours?

- A** \$0.24 **B** \$0.48 **C** \$0.72 **D** \$1.44

38. M/J/13/12/Q36

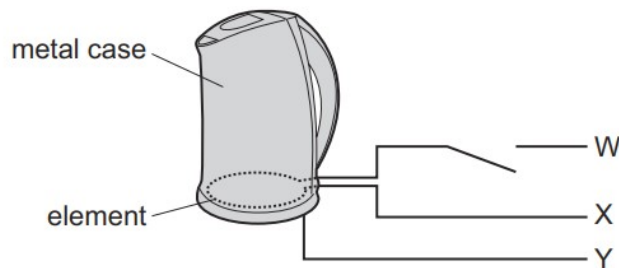
Five electrical appliances are connected to the same socket and there is a very large current.

Why is this dangerous?

- A** The fuses blow in the appliances.
B There is a greater risk of an electrical shock.
C There is overheating in each appliance.
D There is overheating in the socket.

39. M/J/13/11/Q33

The diagram represents part of a household circuit containing an electric kettle.



Which row correctly identifies the wires W, X and Y?

	W	X	Y
A	earth	live	neutral
B	live	neutral	earth
C	live	earth	neutral
D	neutral	live	earth

40. O/N/12/11/Q31

A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

- A** 30 J **B** 36 J **C** 120 J **D** 360 J

41. O/N/12/11/Q32

An immersion heater is labelled 12 V, 60 W.

What is the current in the heater when connected to a 12 V supply?

- A** 0.20 A **B** 5.0 A **C** 12 A **D** 60 A

42. O/N/12/12/Q31

A lamp is rated at 60 W on a 240 V supply.

What is the current in the lamp when used normally?

- A** 0.25 A **B** 4.0 A **C** 60 A **D** 180 A

43. O/N/12/12/Q32

The cost of electricity is 16 c per kWh.

What is the cost of using a 3 kW heater for 4 hours?

- A** $\frac{3 \times 4}{16}$ c **B** $\frac{3 \times 16}{4}$ c **C** $4 \times 3 \times 16$ c **D** $4 \times 3 \times 60 \times 16$ c

44. M/J/12/12/Q31

A lamp is rated at 12 V, 600 mW.

What is the current in the lamp?

- A** 20 mA **B** 50 mA **C** 2.0 A **D** 5.0 A

45. M/J/12/12/Q32

A water heater uses 6 kW of electric power when connected to a 240 V circuit.

Which fuse is **most** suitable for use in this circuit?

- A** 5 A **B** 13 A **C** 30 A **D** 50 A

46. M/J/12/11/Q31

A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

- A** 30 J **B** 36 J **C** 120 J **D** 360 J

47. M/J/12/11/Q32

What causes the fuse to blow in a mains electrical circuit?

- A** a person touches the live wire
B a person touches the neutral wire
C the live wire touches the earth wire
D the neutral wire touches the earth wire

48. O/N/11/11/Q30

The current in a filament lamp is 0.25 A when working normally. The lamp is connected to a plug and the mains a.c. supply.

When the lamp is switched on, it does not light.

What is a possible cause for this?

- A** The earth wire in the plug is not connected.
- B** The fuse in the plug is 3 A .
- C** The lamp only works on a d.c. power supply.
- D** The live wire in the plug is not connected.

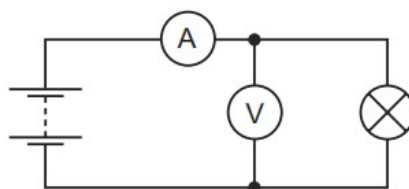
49. O/N/11/11/Q31

Which appliance used on a 240 V supply is **most** likely to burn out its fuse?

	appliance	fuse
A	100 W lamp	1 A
B	1 kW vacuum cleaner	5 A
C	2 kW fan heater	3 A
D	3 kW electric fire	13 A

50. M/J/11/12/Q32

The diagram shows a circuit.



The lamp is a 12 W lamp and is working at normal brightness.

What are the readings on the meters?

	voltmeter reading / V	ammeter reading / A
A	6	0.5
B	12	0.5
C	12	1
D	24	2

51. O/N/10/11/Q31

Many electrical appliances have metal cases.

To prevent the case from becoming 'live', with the possibility of an electric shock, the earth wire of the electric cable is attached to the case.

How does the earth wire prevent an electric shock?

- A** It allows a current to flow to earth, so that the appliance continues working.
- B** It allows a large current to flow to earth, blowing the fuse.
- C** It prevents the fuse from blowing.
- D** It reduces the current to a safe level.

52. M/J/10/12/Q29

The metal case of an electric heater is earthed. The plug to the heater contains a 5 A fuse. There is a current of 4 A when the heater works normally.

The cable to the heater becomes so worn that the live wire makes electrical contact with the case.

What happens?

- A** The current flows to earth and the fuse is not affected.
- B** The fuse melts and switches off the circuit.
- C** The metal case becomes live and dangerous.
- D** The metal case becomes very hot.

53. O/N/09/Q30

Which costs the **most** if operated from the same mains supply?

- A** a 5000 W electric cooker used for 1 minute
- B** a 1000 W electric fire used for 10 minutes
- C** a 500 W electric iron used for 1 hour
- D** a 100 W lamp used for 1 day

54. M/J/09/Q29

An electric iron is marked 240 V, 2500 W.

Four fuses are available with values of 5 A, 10 A, 13 A and 30 A.

Which fuse should be used?

- A** 5 A **B** 10 A **C** 13 A **D** 30 A

55. M/J/09/Q28

A student has a chain of 20 lamps. These are wired in series and connected to the mains.

One lamp blows and all the others go out.

The student wants to find the faulty lamp and replace it.

Where should the student begin?

- A** anywhere, because the current was the same in each lamp
- B** at the live end of the chain, because the current was greatest there
- C** at the middle of the chain, because the current was greatest there
- D** at the neutral end of the chain, because the current was least there

56. O/N/08/Q31

An electric heater is rated at 3 kW. Electrical energy costs 20 cents per kWh.

What is the cost of using the heater for five hours?

- A** 12 cents **B** 60 cents **C** 100 cents **D** 300 cents

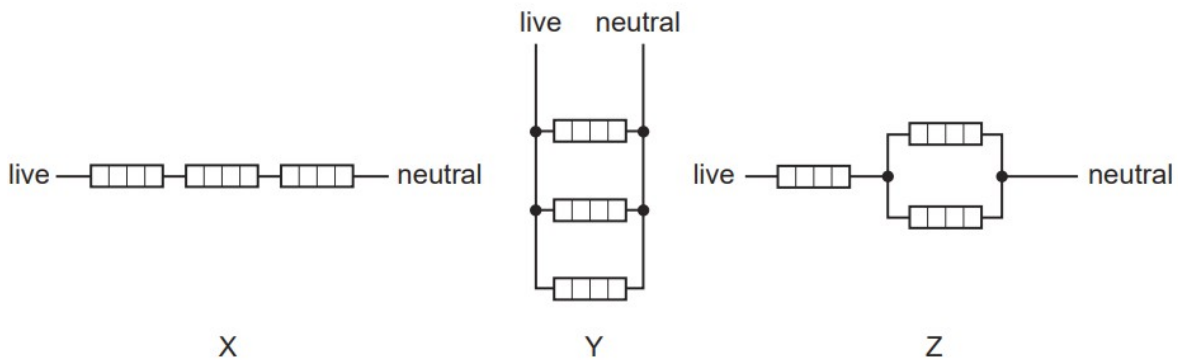
57. O/N/08/Q32

What is the purpose of a circuit breaker in an electric circuit?

- A** to change alternating current into direct current
- B** to keep the current constant
- C** to prevent the current from becoming too large
- D** to reduce the current to a safe value

58. M/J/08/Q30

Three identical heating elements are wired up to the mains supply in the three arrangements shown.



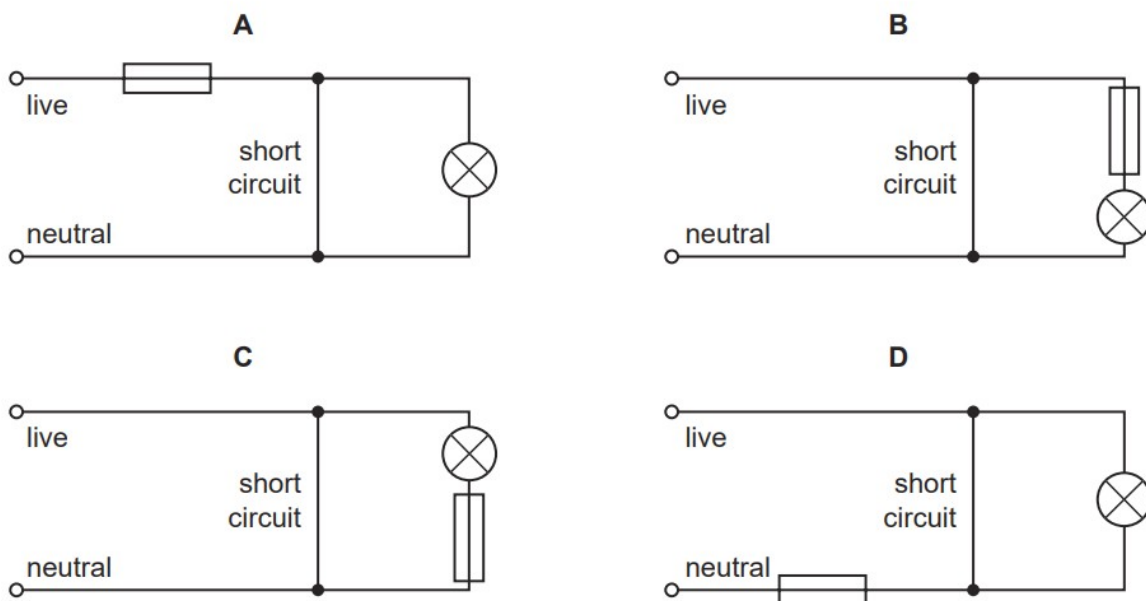
In which arrangement is the current from the supply lowest and in which is it highest?

	lowest current	highest current
A	X	Z
B	X	Y
C	Y	X
D	Y	Z

59. M/J/08/Q32

In each of the circuits below there is a short circuit.

In which circuit does the fuse blow and make the circuit safe to repair?



60. O/N/07/Q31

Energy is represented by the letter E , current by I , power by P , charge by Q , p.d. by V and time by t .

Which pair of equations is correct?

A $E = It$ and $P = VI$

B $E = VQ/t$ and $P = VI$

C $E = VI$ and $P = VI$

D $E = VQ$ and $P = VI/t$

61. O/N/06/Q30

A house-owner replaces a failed fuse for the lights of the house. When the lights are switched on, the second fuse also fails. The house-owner then uses a third fuse with a higher rating than the previous two.

Why is this **not** a sensible thing to do?

A Fuses only allow the circuit to work if the rating is exactly right.

B The third fuse will melt because the rating is too high.

C Using a fuse with too high a rating causes electric shocks.

D The circuit may work, but the fault is not corrected.

62. O/N/05/Q30

A combined bathroom unit of a heater and a lamp is controlled by one switch. The unit contains a 2 kW heater and a 100 W lamp. In one week, the lamp uses 1 kWh of electrical energy.

How much electrical energy is used by the heater alone?

A 2 kWh

B 4 kWh

C 10 kWh

D 20 kWh

63. O/N/05/Q31

An electric kettle is plugged in and switched on. The fuse in the plug blows immediately.

Which single fault could cause this?

A The earth wire is not connected to the kettle.

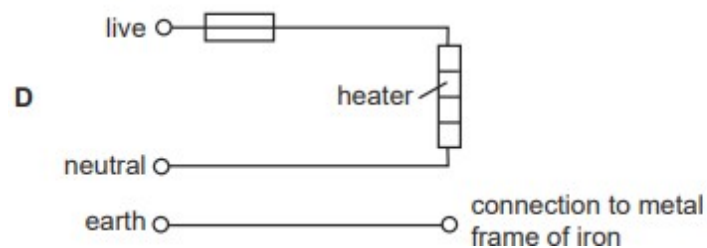
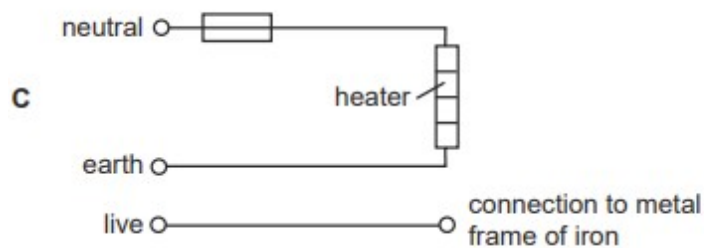
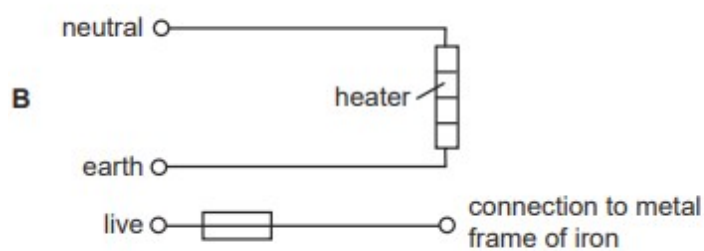
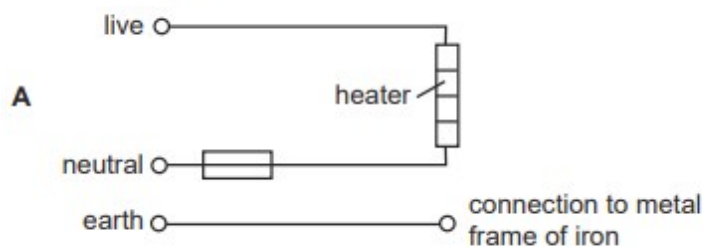
B The live wire and neutral wire connections in the plug are swapped around.

C The live wire touches the metal case of the kettle.

D The wires connected to the plug are too thin.

64. O/N/04/Q32

Which circuit shows how the heater in an electric iron and a fuse should be connected to a mains electricity supply?



65. M/J/04/Q30

In order to turn off both lamps, which is the safest switch position?

